

Solution

GRAVITY-03

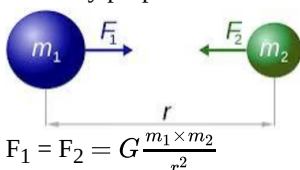
Class 09 - Science

Section A

1. The correct order of match is given as (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i).
2. The correct order of match is given as (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv).
3. The correct order of match is given as (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii).
4. The correct order of match is given as (a) – (ii), (b) – (iv), (c) – (ii), (d) – (i).
5. The correct order of match is given as (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i).
6. The correct order of match is given as (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii).
7. The correct order of match is given as (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii).
8. The correct order of match is given as (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii).

Section B

9. It is called force of gravity.
10. When we weigh our body, an upward force on it. The upward force is the buoyant force. As a result the body gets pushed slightly upwards, causing the weighing machine to show the value less than the actual value.
11. Gravitational force.
12. Gravitational force of earth is the force by which earth exerts on any object towards itself.
Weight is the force which the object exerts on the earth.
13. The average acceleration of free fall on the surface of earth is 9.81 ms^{-2} .
14. We have given
Mass, $m = 10 \text{ kg}$
Acceleration due to gravity, $g = 9.8 \text{ ms}^{-2}$
 $W = m \times g$
 $W = 10 \text{ kg} \times 9.8 \text{ ms}^{-2} = 98 \text{ N}$
Thus, the weight of the object is 98 N.
15. When an object falls with a constant acceleration, under the influence of force of gravitation of the earth, object is said to have free fall.
16. According to Archimedes Principle when a body is partially or completely immersed in a fluid at rest is acted upon by an upward, or buoyant, force the magnitude of which is equal to the weight of the fluid .
17. Everybody in this universe attracts every other body with a force, which is directly proportional to the product of their masses and inversely proportional to the square of distance between their centres.



18. They do not exert any force/weight on their spaceship in the absence of gravity in space.
19. The Earth and the Moon experience equal gravitational forces from each other. However the mass of the Earth is much larger than the mass of the moon. Hence it accelerates at a rate lesser than the acceleration rate of the moon towards the Earth. For this reason Earth does not move towards the moon.
20. Stone falls due to gravitation force exerted by the earth. Earth does not move towards the stone as the mass of earth is much more than that of the stone.
21. The earth does not fall into the sun because the earth remains in its circular orbit due to the gravitational force acting on it.
22. No, his weight will not remain same as that at the poles. There will be a decrease in his weight at the equator. As the radius of the earth increase from the poles to the equator, the value of 'g' becomes greater at poles decreasing towards equator. Also, the force of gravity decrease from poles to the equator.
23. The moon will begin to move in a straight line in the direction in which it was moving at that instant because the circular motion of the moon is due to centripetal force provided by the gravitational force of the earth.
24. When an object is immersed partially or fully in a liquid, it experiences an upward force. This upward force is known as buoyant force and this phenomenon is known as buoyancy. Buoyancy is also known as up thrust

25. Henry Cavendish
26. An object immersed in a liquid experiences buoyance force In the upward direction only.
27. Both will reach the ground at the same time because acceleration due to gravity is independent of the masses of freely falling bodies.
28. It is a state when objects do not weigh anything. This occurs when bodies are in a state of free fall under the effect of gravity.
29. We know that,
 Weight of object on the moon = $(\frac{1}{6}) \times$ its weight on the earth.
 That is,

$$W_m = \frac{W_e}{6} = \frac{10}{6} \text{ N}$$

$$= 1.67 \text{ N}$$
 Thus, the weight of the object on the surface of the moon would be 1.67 N.
30. As the acceleration due to gravity on the moon is one-sixth that on the earth, hence the weight of the man will decrease on the surface of moon by 100/6 i.e. it becomes 16.66.
31. The mass of the two bodies are very small. Therefore, the force of attraction between them is very small. So, both the objects do not move towards each other.
32. i. The mass of the body will be 30Kg because mass remains the same as it is constant.
 ii. The distance of the body from the centre of the earth = 6370 + 230 Km

$$= 6600 \text{ Km} = 6.6 \times 10^6 \text{ m}$$
 Acceleration due to gravity = $g = \frac{Gme}{r^2}$

$$g = \frac{6.673 \times 10^{-11} \times 5.98 \times 10^{24}}{(6.6 \times 10^6)^2}$$

$$g = 9.16 \text{ m/s}^2$$
 iii. Weight at that height = mg

$$= 30 \times 9.16$$

$$= 274.8 \text{ N}$$
33. Since weight of a body on the Earth is given by $W = m \times g$, where m is the mass and g is the acceleration. The value of g is greater at the poles than at the equator, hence weight of same amount of gold will be lesser at equator than it was on the poles. Therefore, the friend will not agree with the weight of gold bought.
34. As we know,

$$F = \frac{Gm_1m_2}{r^2}$$
 [symbols have their usual meanings]
 i. $r' = 3r \Rightarrow F' = \frac{Gm_1m_2}{9r^2} = \frac{F}{9}$ [force decreases by 9 times]
 ii. $m'_1 = 2m_1$ and $m'_2 = 2m_2 \Rightarrow F' = \frac{4Gm_1m_2}{r^2} = 4F$ [force increases by 4 times]
 iii. $m'_1 = 2m_1$ and $m'_2 = 2m_2$ $r' \Rightarrow F' = \frac{4Gm_1m_2}{4r^2} = F$ [force remains unchanged]
35. Given, $M_e = 6 \times 10^{24} \text{ kg}$, $m = 25 \text{ kg}$
 $R_e = 6.4 \times 10^6 \text{ m}$ and $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$
 i. $\therefore$ Mutual force, $F = G \frac{M_e}{R_e^2} m = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 25}{(6.4 \times 10^6)^2} = 244 \text{ N}$
 ii. $\therefore$ Acceleration produced in the body, $a = \frac{F}{m} = \frac{244}{25} = 9.8 \text{ m/s}^2$
 iii. $\therefore$ Acceleration produced in the earth, $a = \frac{F}{M_e} = \frac{244}{6 \times 10^{24}} = 4.06 \times 10^{-23} \text{ m/s}^2$
36. Case (i) = For the first ball
 Initial velocity (u) = 0
 Final velocity (v) = ?
 Acceleration due to gravity (g) = 10 ms^{-2}
 Distance of fall (S) = 20 m
 Time of fall (t) = ?
 Using $S = ut + \frac{1}{2}at^2$ we have time of fall of the first ball

$$20 = 0 \times t + \frac{1}{2} \times 10 \times t^2$$

$$\Rightarrow t^2 = 4$$

$$\Rightarrow t = 2 \text{ s}$$
 Since the second ball is thrown 1 second later. Therefore for the two balls to reach the ground at the same time, the second ball

should in motion for $t = 2 - 1 = 1$ second

Therefore using the equation $S = ut + \frac{1}{2}at^2$ we have

$$20 = u \times 1 + \frac{1}{2} \times 10 \times 1^2$$

$$\Rightarrow u = 15 \text{ m s}^{-1}$$

37. We have given that,

The mass of the earth, $M = 6 \times 10^{24}$ kg

The mass of the moon, $m = 7.4 \times 10^{22}$ kg

The distance between the earth and the moon,

$$d = 3.84 \times 10^5 \text{ km}$$

$$= 3.84 \times 10^5 \times 1000 \text{ m}$$

$$= 3.84 \times 10^8 \text{ m}$$

$$G = 6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

The force exerted by the earth on the moon is

$$F = G \frac{M \times m}{d^2}$$

$$= \frac{6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2} \times 6 \times 10^{24} \text{ kg} \times 7.4 \times 10^{22} \text{ kg}}{(3.84 \times 10^8 \text{ m})^2}$$

$$= 2.02 \times 10^{20} \text{ N}$$

Thus, the force exerted by the earth on the moon is 2.02×10^{20} N.

38. Let the initial velocity = u

Let the maximum height reached = h m

When it reached, the velocity = 10 m/s

$$\text{Now } v^2 = u^2 + 2gh$$

$$(10)^2 = (u)^2 + 2(-10) \times \frac{h}{2}$$

$$(u)^2 = 100 + h \times 10$$

When the ball reaches the highest point, $v = 0$

$$v^2 - u^2 = 2gh$$

$$0^2 - u^2 = 2(-10)h$$

$$-u^2 = -2(10)h$$

$$\Rightarrow (100 + 10h) = 20h$$

$$100 = 20h - 10h$$

$$100 = 10h$$

$$\text{Height} = 10 \text{ m}$$

Maximum height reached = 10 m

$$u^2 = 100 + 10h$$

$$u^2 = 100 + 10 \times 10$$

$$u^2 = 200$$

$$u = \sqrt{200} = 14.14 \text{ m/s}$$

Initial velocity of the ball when it was thrown up = 14.14 m/s .

Velocity after 1 s

$$v = u + at$$

$$v = 14.14 + (-10) \times 1$$

$$= 4.14 \text{ m/s}$$

$$\text{Acceleration after } 1 \text{ s} = -10 \text{ m/s}^2$$

39. i. No, her friend will not agree with the weight of gold bought because weight at poles is greater than the weight at equator.

ii. We know that the gravitational force is always attractive, still, the moon does not fall on the earth because the gravitational force between the earth and the moon works as the necessary centripetal force for the moon to make it revolve around the earth.

$$40. \therefore \text{Mass of cuboid} = \text{Volume} \times \text{Density} = (4 \text{ cm} \times 7 \text{ cm} \times 10 \text{ cm}) \times 8 \text{ g cm}^{-3} [\because \text{Volume of cuboid} = l \times b \times h]$$
$$= 2240 \text{ g} = 2.24 \text{ kg}$$

$$\text{Force applied by the cuboid on the sand} = mg = 2.24 \times 10 = 22.4 \text{ N} [\because g = 10 \text{ ms}^{-2}]$$

Pressure will be minimum when area of the face of cuboid kept on sand is maximum, i.e. in the case of face with $7 \text{ cm} \times 10 \text{ cm}$.

$$\text{Area of the race} = 7 \text{ cm} \times 10 \text{ cm} = \frac{7}{100} \text{ m} \times \frac{10}{100} \text{ m}$$

$$= 0.07 \text{ m} \times 0.1 \text{ m} = 0.007 \text{ m}^2$$

$$\text{Minimum pressure} = \frac{\text{Force}}{\text{Area}} = \frac{22.4}{0.007} = 3200 \text{ Nm}^{-2}$$

Pressure will be maximum when area of face of cuboid kept on the sand is minimum, i.e. in the case of face with $4 \text{ cm} \times 7 \text{ cm}$.

$$\text{Area of the face} = 4 \text{ cm} \times 7 \text{ cm} = \frac{4}{100} \text{ m} \times \frac{7}{100} \text{ m} = 0.0028 \text{ m}^2$$

$$\text{Maximum pressure} = \frac{22.4}{0.0028} = 8000 \text{ Nm}^{-2}$$

41. i. Gravitational force is the weakest force in nature, yet it helps in holding massive bodies in place throughout the universe. Due to this reason, table and chair do not collide.

ii. Given, $h = 12 \text{ m}$, $v = 0$, $g = -9.8 \text{ m/s}^2$, $u = ?$ and $t = ?$

$$\text{a. } \therefore v^2 - u^2 = 2gh, 0 - u^2 = 2 \times (-9.8) \times 12$$

$$\Rightarrow u = 15.33 \text{ m/s}$$

$$\text{b. } \therefore v = u - gt$$

$$\Rightarrow t = \frac{u}{g} = \frac{15.33}{9.8} = 1.56 \text{ s}$$

42. Height of coconut (S) = 15 m

$$\text{Acceleration due to gravity (g)} = -10 \text{ ms}^{-2}$$

$$\text{Time (t)} = ?$$

$$\text{Initial velocity (u)} = 20 \text{ ms}^{-1}$$

$$\text{Using } S = ut + \frac{1}{2}at^2 \text{ we have}$$

$$15 = 20 \times t + \frac{1}{2} \times (-10)t^2. \text{ This is a quadratic equation in time.}$$

Simplifying we have

$$5t^2 - 20t + 15 = 0 \text{ or } t^2 - 4t + 3 = 0$$

$$t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times 3}}{2} = \frac{4 \pm 2}{2} = 1 \text{ or } 3 \text{ s} = 1 \text{ or } 3 \text{ s}$$

Thus the projectile will pass the coconut 1s after it is thrown while moving in the upward direction and 3 s while moving in the downward direction on its return journey.

43. $\therefore$ The ball returns back to the thrower in 6s, the time for its upward journey = $6/2 = 3 \text{ s}$

For the upward motion of the ball

$$\text{Initial velocity (u)} = ?$$

$$\text{Final velocity (v)} = 0 (\because \text{Ball comes to rest})$$

$$\text{Time (t)} = 3 \text{ s}$$

$$\text{Acceleration due to gravity (g)} = -10 \text{ ms}^{-2} [\text{In upward direction g is -ve}]$$

i. We know:

$$v = u + gt$$

$$\Rightarrow 0 = u - 10 \times 3$$

$$\Rightarrow -u = -30$$

$$\Rightarrow u = 30 \text{ ms}^{-1}$$

ii. We know:

$$S = ut + \frac{1}{2}gt^2$$

$$S = 30 \times 3 - \frac{1}{2} \times 10 \times (3)^2$$

$$\Rightarrow S = 90 - 45$$

$$S = 45 \text{ m}$$

iii. For the downward motion of ball

$$\text{Initial velocity (u)} = 0$$

$$\text{Time for downward fall (t)} = 4 - 3 = 1 \text{ s}$$

$$\text{Acceleration due to gravity (g)} = 10 \text{ ms}^{-1}$$

$$\text{Distance covered in downward direction (S)} = ?$$

We know:

$$S = ut + \frac{1}{2}gt^2$$

$$S = 0 + t + \frac{1}{2} \times 10 \times (1)^2$$

$$\Rightarrow S = 0 + 5$$

$$\Rightarrow S = 5 \text{ m}$$

Position of ball after 4s from ground

$$= 45 - 5$$

$$= 40 \text{ m.}$$

$$44. (a) \frac{\text{The Density of the floating body}}{\text{Density of fluid in which it floats}} = \frac{\text{Height immersed}}{\text{Total height}}$$

$$\frac{0.8}{1.0} = \frac{h}{4.2}$$

$$h = 4.2 \times 0.8 = 3.36 \text{ m}$$

b) Mass of the block of wood = 100 Kg

Mass of water displaced = 100 Kg

Mass of lead placed on the block of wood = 10 Kg

Total mass of the block of wood + weight placed on it = 110 Kg

Mass of water now displaced = 110 Kg

When 100 Kg of water was displaced, 3.36 m was submerged

When 110 Kg of water is to be displaced,

$$\frac{110 \times 3.36}{100} \text{ m will be submerged.}$$

$$= 3.696 \text{ m will be submerged.}$$

$$\text{Height seen above water} = 4.2 - 3.696$$

$$= 0.504 \text{ m}$$

45. As $u = 0$, using equation, $S = ut + \frac{1}{2}at^2$, We have,

$$h_1 = \frac{1}{2}gt_1^2$$

$$\Rightarrow h_2 = \frac{1}{2}gt_2^2,$$

$$\Rightarrow \frac{t_1}{t_2} = \sqrt{\frac{h_1}{h_2}}$$

Since, time period does not depend upon the mass and size. Therefore, Ratio will not change in either case because acceleration remains the same.

46. i) The upthrust on cork due to water = Weight of water displaced by the cork

= Weight of cork in air - Weight of cork in water

$$= (W_2 - W_1) - (W_3 - W_1) = (W_2 - W_3)$$

$$= 220 - 140$$

$$= 80 \text{ gf}$$

ii) Since the density of water is 1 g cm^{-3} , therefore volume of water displaced = 80 cm^3

Hence volume of cork = 80 cm^3

$$\text{iii) RD of cork} = \frac{\text{weight of cork in air}}{\text{weight of water displaced by cork}}$$

$$= \frac{W_2 - W_1}{W_2 - W_3} = \frac{220 - 200}{220 - 140} = \frac{20}{80} = 0.25$$

47. Given that mass of Jupiter, $M_1 = 1.9 \times 10^{27} \text{ kg}$

$$\text{mass of sun, } M_2 = 1.99 \times 10^{30} \text{ kg}$$

Distance between Jupiter and sun, $r = 7.8 \times 10^{11} \text{ m}$ and

$$G = 6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$$

Using $F = \frac{GM_1M_2}{r^2}$ we have

$$F = \frac{6.67 \times 10^{-11} \times 1.9 \times 10^{27} \times 1.99 \times 10^{30}}{(7.8 \times 10^{11})^2}$$

$$= 4.416 \times 10^{23} \text{ N}$$

48. According to the question

$$M_{\text{sun}} = \text{Mass of the sun} = 2 \times 10^{30} \text{ Kg}$$

$$M_{\text{Earth}} = \text{Mass of the Earth} = 6 \times 10^{24} \text{ Kg}$$

R is the average distance between the sun and the Earth = $1.5 \times 10^{11} \text{ m}$

$$F = G \frac{Mm}{d^2}$$

$$F = \frac{6.7 \times 10^{-11} \times 2 \times 10^{30} \times 6 \times 10^{24}}{(1.5 \times 10^{11})^2}$$

$$= 35.73 \times 10^{21} \text{ N}$$

49. i. Time taken by ball to reach maximum height(t) = 6/2 = 3 s

$$v = u + gt$$

$$0 = u + (-9.8) \times 3$$

$$u = 29.4 \text{ m/s (the velocity with which it was thrown up)}$$

ii. The maximum height it reaches: therefore $h = ut + \frac{1}{2}gt^2$

$$= 29.4 \times 3 + \frac{1}{2} \times (-9.8) \times 3^2$$

$$= 88.2 - 44.1$$

$$= 44.1 \text{ m}$$

iii. Its position after 4 s will be:

Since in first 3 s it will reach the maximum height and in next 1 s it will start a free fall so, $u = 0$, $t = 1$

$$h = ut + \frac{1}{2}gt^2$$

$$= 0 \times t + \frac{1}{2} \times 9.8 \times 1 = 4.9 \text{ m}$$

therefore, after 4s the position of ball = 44.1 - 4.9 = 39.2 m

50. Let m = Mass of the body

M_e = Mass of earth

M_m = Mass of Mars

r_e = Radius of earth

r_m = Radius of Mars

G = Universal gravitational constant

Weight of body on earth = F_e

$$= \frac{GM_em}{(r_e)^2}$$

$$\text{Weight of body on Mars} = F_m = \frac{GM_mm}{(r_e)^2}$$

Divide one equal by another

$$\frac{F_m}{F_e} = \frac{GM_mm \times (r_e)^2}{(r_m)^2 \times GM_em}$$

$$\frac{F_m}{F_e} = \frac{M_m(r_e)^2}{M_e(r_m)^2}$$

$$\text{As, } M_m = 0.11 M_e$$

$$F_e = 200 \text{ N}$$

$$r_e = 6370 \text{ Km}$$

$$r_m = 3400 \text{ Km}$$

$$\frac{F_m}{200} = \frac{0.11 M_e \times (6.37 \times 10^6)^2}{M_e \times (3.4 \times 10^6)^2}$$

$$F_m = \frac{0.11 \times (6.37 \times 10^6)^2 \times 200}{(3.4 \times 10^6)^2}$$

$$F_m = 77.22 \text{ N}$$

Weight of the body on Mars = 77.22N

51. The up thrust = weight of the air displaced

$$\text{Mass of air displaced} = 120 \times 1.14$$

$$= 136.8 \text{ Kg}$$

$$\text{Weight of air displaced} = 136.8 \times 9.8 \text{ N} = (Mg)$$

$$= 1340.6 \text{ N}$$

Downward force = weight of the balloon + weight of H_2

Now, Mass of the balloon + Mass of H_2

$$= 75 + 120 (0.081)$$

$$= 84.72 \text{ Kg}$$

Weight of the balloon and H_2 filling it = 84.72×9.8

$$= 830.3 \text{ N}$$

$$\text{Net up thrust} = 1340.6 - 830.3$$

$$= 510.3\text{N}$$

Maximum weight the balloon can lift = 510.3N

52. Given, the mass of the mars, $M_m = 6.39 \times 10^{23} \text{ kg}$

The mass of the Jupiter, $M_j = 1.89 \times 10^{27} \text{ kg}$

The distance between the mars and jupiter,

$$d = 7.49 \times 10^5 \text{ m}$$

$$G = 6.7 \times 10^{-11} \text{ Nm}^2 / \text{Kg}^2$$

The force exerted by the Jupiter on the mars,

$$F = G \frac{M_m \times M_j}{d^2} = \frac{6.7 \times 10^{-11} \times 6.39 \times 10^{23} \times 1.89 \times 10^{27}}{(7.49 \times 10^5)^2} = 1.44 \times 10^{29} \text{ N}$$

Thus, the force exerted by the Jupiter on the mars is $1.44 \times 10^{29} \text{ N}$.

53. For the first stone, Initial velocity $u = 0$

Let the stone take t s to reach the ground

$$s = ut + \frac{1}{2}gt^2$$

$$-49 = \frac{1}{2}(-9.8)t^2$$

$$t^2 = 10$$

$$t = 3.162\text{s}$$

For the second stone, the initial velocity = u_0

Time of flight = 3.162-1 s

$$= 2.162\text{s}$$

$$s = ut + \frac{1}{2}gt^2$$

$$-49 = -u_0(2.162) + \frac{1}{2}(-9.8)(2.162)^2$$

$$-49 = -2.16u_0 - 22.9$$

$$u_0 = \frac{26.1}{2.162} = 12.1\text{m/s}$$

The second stone was thrown downward with a velocity of 12.1 m/s

54. The volume of the block = $(0.2 \times 0.2 \times 0.4) \text{ m}^3$

$$= 0.016 \text{ m}^3$$

Mass of the block = $0.016 \times 600 = \text{Density} \times \text{Volume}$

$$= 9.6 \text{ Kg}$$

Weight of the block = $M \times g$ ($M = \text{Mass}$, $g = \text{Acceleration due to gravity}$)

$$= 9.6 \text{ g}$$

Up thrust on the block = Weight of water displaced

Mass of water displaced = volume of the block $\times$ Density of water

$$= 0.016 \times 1000$$

$$= 16\text{Kg}$$

Weight of the water displaced = Mass $\times$ acceleration due to gravity

$$= 16 \times g$$

$$= 16g$$

$$\text{Up thrust} = 16g \text{ N}$$

$$\text{Weight of the block} = 9.6g\text{N}$$

$$\text{Net up thrust} = 16g - 9.6g$$

$$= 6.4g\text{N}$$

$$= 6.4 \times 9.8 \text{ N}$$

$$= 62.72 \text{ N}$$

The tension in the string = Net up thrust

$$= 62.72\text{N}$$

55. $v = \text{Initial velocity}$

$s = \text{Distance}$

$t = \text{Time}$

$g = \text{Acceleration due to gravity}$

Let two stones meet after t s.

$$s = ut + \frac{1}{2}gt^2$$

$$u = 0$$

$$s = \frac{1}{2}(-10)t^2$$

Magnitude of height through which the stone falls = $5t^2$

Height reached by the stone moving up is t s

$$s = ut + \frac{1}{2}gt^2$$

$$s = 50t + \frac{1}{2}(-10)t^2$$

$$s = 50t - 5t^2$$

Total distance travelled by 2 stones = 50m

$$5t^2 + 50t - 5t^2 = 50$$

$$t = 1$$

The stones meet 1 s after they are dropped or thrown up

The height through which the stone moves up is 1s

$$S = 50t - 5t^2$$

$$= 50 - 5$$

$$= 45 \text{ m}$$

Therefore, the stones meet 45m above the ground.

$$56. \text{ Volume of the cube} = (\text{side})^3 = 5^3 \text{ cm}^3 = 125 \text{ cm}^3$$

$$= 125 \times 10^{-6} \text{ m}^3$$

$$\text{Volume of ethanol displaced} = 1.25 \times 10^{-4} \text{ m}^3$$

$$\text{Density of ethanol} = 0.8 \text{ g/cm}^3$$

$$1 \text{ cm}^3 \text{ of ethanol has a mass of } 0.8$$

$$1 \text{ m}^3 \text{ (that is } 10^6 \text{ cm}^3) \text{ of ethanol will have a mass} = 0.8 \times 10^6 \text{ g}$$

$$= 8 \times 10^5 \text{ g} = 800 \text{ Kg}$$

$$\text{The Density of ethanol} = 800 \text{ Kg/m}^3$$

$$\text{Mass of ethanol} = 1.25 \times 10^{-4} \times 800 = 0.1 \text{ Kg}$$

$$\text{Weight of ethanol displaced} = 0.1 \times 9.8 = 0.98 \text{ N}$$

By Archimedes's principles

$$\text{Up thrust} = \text{weight of fluid displaced} = 0.98 \text{ N}$$

57. We have given that,

$$\text{Time taken, } t = \frac{1}{2} \text{ second}$$

$$\text{Initial velocity, } u = 0 \text{ ms}^{-1}$$

$$\text{Acceleration due to gravity, } g = 10 \text{ ms}^{-2}$$

$$\text{Acceleration of the car, } a = + 10 \text{ ms}^{-2} \text{ (downward)}$$

i. speed $v = at$

$$v = 10 \text{ ms}^{-2} \times 0.5 \text{ s}$$

$$= 5 \text{ ms}^{-1}$$

Thus,

$$\text{Its speed on striking the ground} = 5 \text{ ms}^{-1}$$

$$\text{ii. Average speed} = \frac{u+v}{2}$$

$$= \frac{(0 \text{ ms}^{-1} + 5 \text{ ms}^{-1})}{2}$$

$$= 2.5 \text{ ms}^{-1}$$

Thus,

$$\text{Its average speed during the } 0.5 \text{ s} = 2.5 \text{ ms}^{-1}$$

$$\text{iii. Distance travelled, } s = \frac{1}{2}at^2$$

$$= \frac{1}{2} \times 10 \text{ ms}^{-2} \times (0.5 \text{ s})^2$$

$$= \frac{1}{2} \times 10 \text{ ms}^{-2} \times 0.25 \text{ s}^2$$

$$= 1.25 \text{ m}$$

Thus,

Height of the ledge from the ground = 1.25 m

58. **Acceleration due to gravity:** The acceleration produced in the motion of a body falling under the force of gravity is called acceleration due to gravity. It is denoted by 'g'. It is expressed in units, ms^{-2} .

Expression for acceleration due to gravity: The force (F) of gravitational attraction on a body of mass m due to earth of mass M and radius R is given by, $F = G \frac{mM}{R^2}$ (1)

Where, 'G' is universal gravitational constant.

According to Newton's second law of motion: Force is the product of mass and acceleration.

$\therefore F = ma$

But the acceleration due to gravity is represented by the symbol g.

Therefore, we can write; $F = mg$ (2)

From equation (1) and (2), we get

$$mg = G \frac{mM}{R^2} \text{ or } g = \frac{GM}{R^2} \dots(3)$$

When the body is at a distance 'R' from centre of the earth then $g = \frac{GM}{R^2}$.

It may be noted that, value of 'g' is independent of mass of object.

59. i. We know that, Original weight, $W_o = mg = \frac{GMm}{R^2}$, where M is the mass of the earth, m = mass of body.

Let the new mass of earth = M'

According to question, New mass, $M' = M + 10\% \text{ of } M = M + \frac{10}{100}M = M + \frac{M}{10} = \frac{11M}{10} = 1.1M$

$$\therefore \text{New weight, } W_n = \frac{GM'm}{R^2} = \frac{G \times 1.1Mm}{R^2}$$

$$\text{Now, Ratio of new weight to original weight} = \frac{\text{New weight}}{\text{Original weight}} = \frac{1.1GMm/R^2}{GMm/R^2} = 1.1$$

New weight becomes 1.1 times the original weight of body.

i.e., weight of body will increase by 10%.

ii. Again, Original Weight, $W_o = \frac{GMm}{R^2}$, where R is the radius of the earth.

According to question, when R changes to 2R, the new weight is given by,

$$\text{New weight, } W_n = \frac{GMm}{4R^2}$$

$$\text{Now, Ratio of new weight to original weight} = \frac{\text{New weight}}{\text{Original weight}} = \frac{GMm/4R^2}{GMm/R^2} = \frac{1}{4}$$

Therefore, New weight becomes $\frac{1}{4}$ times of original weight

60. We have given that,

The mass of the wooden block = 5 kg

The dimensions = 40 cm $\times$ 20 cm $\times$ 10 cm

Here, the weight of the wooden block applies a thrust on the table top.

i.e,

Now we know that,

$$\text{Thrust} = F = m \times g$$

$$= 5 \text{ kg} \times 9.8 \text{ ms}^{-2}$$

$$= 49 \text{ N}$$

Area of a side = length $\times$ breadth

$$= 20 \text{ cm} \times 10 \text{ cm}$$

$$= 200 \text{ cm}^2 = 0.02 \text{ m}^2$$

$$\text{From equation Pressure} = \frac{\text{thrust}}{\text{area}} \dots\dots(i)$$

$$\text{Pressure} = \frac{49 \text{ N}}{0.02 \text{ m}^2}$$

$$= 2450 \text{ Nm}^{-2}.$$

When the block lies on its side of dimensions 40 cm $\times$ 20 cm, it exerts the same thrust.

Area length $\times$ breadth

$$= 40 \text{ cm} \times 20 \text{ cm}$$

$$= 800 \text{ cm}^2 = 0.08 \text{ m}^2$$

From equation (i)

$$\text{Pressure} = \frac{49\text{N}}{0.08\text{m}^2}$$

The pressure exerted by the side $20\text{ cm} \times 10\text{ cm}$ is 2450 Nm^{-2} and by the side $40\text{ cm} \times 20\text{ cm}$ is 612.5 Nm^{-2} .

61. Initial velocity, $u=0$

Total height, $h = 100\text{ m}$

a. Let, for the first 50 m the time taken by the stone be ' t ' sec.

$S=-50\text{ m}$ (- ve sign shows the stone falls in downward direction)

$$g = -10\text{ m/s}^2$$

$$h = s = ut + \frac{1}{2}gt_1^2$$

$$\Rightarrow -50 = 0 + \frac{1}{2}(-10)t_1^2$$

$$\Rightarrow -50 = -5t_1^2$$

$$\Rightarrow \frac{50}{5} = t_1^2$$

$$\Rightarrow t_1^2 = 10$$

$$\Rightarrow t_1 = \sqrt{10}$$

$$\therefore t_1 = 3.16\text{ sec}$$

b. For the entire journey, let the time taken be T

$$u = 0$$

$$S = -100\text{ m}$$

$$a = -10\text{ m/s}^2$$

$$S = ut + \frac{1}{2}aT^2$$

$$\Rightarrow -100 = 0 + \frac{1}{2} \times (-10)T^2$$

$$\Rightarrow T_2 = 20$$

$$\Rightarrow T = \sqrt{20}$$

$$\Rightarrow T = 4.47\text{ sec}$$

$$\therefore \text{Time taken to fall through the next } 50\text{ m} = T - t_1 = 4.47 - 3.16 = 1.31\text{ sec}$$

62. i. Ship displaces more water than needle as the volume of the ship is more than that of the needle. Since upthrust depends on the volume of the object ($U = Vdg$), so more the volume of the object, more upthrust act on it and object floats.

ii. Since, pressure act on the body is inversely proportional to the surface area of contact, i.e.

$$P \propto \frac{1}{A}$$

It means that more the area of contact, less pressure will act on the body. As the broad and the thick handle of our suitcase has a large area, due to which less pressure acts on our hand and it is very easy to take from one place to another.

63.	Mass	Weight
	Mass is a property of matter. The mass of an object is the same everywhere,	Weight depends on the effect of gravity. Weight varies according to location.
	Mass can never be zero.	Weight can be zero if no gravity acts upon an object, as in space.
	Mass does not change according to location.	Weight increases or decreases with higher or lower gravity.
	Mass is a scalar quantity. It has magnitude.	Weight is a vector quantity. It has magnitude and is directed toward the center of the Earth or other gravity well.
	Mass may be measured using an ordinary balance.	Weight is measured using a spring balance.
	Mass usually is measured in grams and kilograms.	Weight often is measured in newtons, a unit of force.

64. For the first stone :

Initial velocity, $u = 0\text{ ms}^{-1}$, Height of cliff, $h = 49\text{ m}$, $g = 9.8\text{ m/s}^2$

As we know, $S = ut + \frac{1}{2}at^2$.

We, have, $h = ut + \frac{1}{2}gt^2$

$$\therefore 49 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2$$

$$\Rightarrow t^2 = \frac{9.8}{9.8} = 10$$

$$\Rightarrow t = \sqrt{10} = 3.16\text{ s}$$

i.e., first stone would take 3.16 s to reach the ground.

For the second stone:

The time taken by the second stone to reach the ground is one second less than that taken by the first stone as both the stones reach the ground from the same height, $h = 49\text{m}$.

That is, for the second stone, time, $t = (3.16 - 1)\text{ s} = 2.16\text{ s}$

∴ For the second stone,

$$g = 9.8\text{ ms}^{-2}, h = 49\text{ m}, t = 2.16\text{ s}, u = ?$$

$$\text{Using, } S = ut + \frac{1}{2}at^2.$$

$$\text{We have, } h = ut + \frac{1}{2}gt^2$$

$$\Rightarrow 49 = u \times 2.16 + \frac{1}{2} \times 9.8 \times (2.16)^2$$

$$\Rightarrow 49 - 22.86 = 2.16u \text{ or } 26.14 = 2.16u$$

$$\Rightarrow u = \frac{26.14}{2.16} = 12.1\text{ ms}^{-1}$$

Therefore, the speed with which he threw the second stone = 12.1 ms^{-1}

65. The value of g on the earth is 9.8 m/s^2

i. a. g on the moon is given by

$$g' = \frac{g}{6} = \frac{9.8}{6} = 1.63\text{ m/s}^2$$

b. Mass of the person on the moon = $\frac{110.84}{1.63} = 68\text{ kg}$

c. Mass will be constant and does not change from place to place. Hence the mass of the person on the earth is the same that on the moon.

$$\text{Weight of person on the earth} = mg = 68 \times 9.8 = 666.4\text{ N}$$

ii. According to the Newton's law of gravitation, the force of attraction between earth and the body is given by

$$F = \frac{GMm}{R^2} \dots(i)$$

where, M = mass of the earth, R = radius of the earth, m = mass of person and $G = 6.67 \times 10^{-11}\text{ N-m}^2/\text{kg}^2$

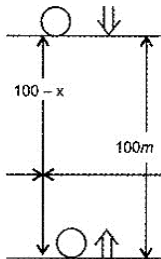
Force produces an acceleration ' g '. So from Newton's second law, $F = mg \dots(ii)$

Equating (i) and (ii) we get,

$$mg = \frac{GMm}{R^2}$$

$$\therefore g = \frac{GM}{R^2}$$

66. Acceleration due to gravity (g) = 10 ms^{-2}



Initial velocity (u) = 0

Distance (S) = $100 - x$

Time (t) = ?

$$S = ut + \frac{1}{2}gt^2$$

$$\Rightarrow (100 - x) = 0 \times t + \frac{1}{2}10 \times t^2$$

$$\Rightarrow 100 - x = 5t^2 \dots(1)$$

For the stone moving vertically upward:

Initial velocity (u) = 25 ms^{-1}

Time (t) = ?

Acceleration due to gravity (g) = -10 ms^{-2}

[In upward direction g is -ve]

Distance (S) = x

$$\text{We know: } S = ut + \frac{1}{2}gt^2$$

$$\Rightarrow x = 25 \times t + \frac{1}{2}(-10t^2)$$

$$\Rightarrow x = 25t - 5t^2 \dots(2)$$

Substituting the value of x from (2) in (1) we get,

$$100 - (25t - 5t^2) = 5t^2$$

$$100 - 25t + 5t^2 = 5t^2$$

$$25t = 100$$

$$t = 4 \text{ s}$$

Put the value of t in (1)

$$\Rightarrow 100 - x = 5(4)^2$$

$$\Rightarrow 100 - x = 80$$

$$x = 20 \text{ m}$$

$\therefore$ the stones will meet at a height of 20 m from ground, after 4s.

67. i. Formula to find the magnitude of gravitational force:

$$F = \frac{GMm}{R^2}.$$

where, M = mass of the earth

m = mass of the object

R = distance between centres of the earth and an object.

and universal gravitational constant, $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$

- ii. a. Let gravitational force be F when the distance between them is R,

$$F = \frac{GMm}{R^2} \dots(i)$$

Now, when the distance reduces to half,

$$F' = \frac{GMm}{\left(\frac{R}{2}\right)^2} = \frac{4GMm}{R^2} = 4F$$

i.e. the force of gravitation becomes 4 times the original value.

- b. When the mass becomes 4 times,

$$F' = \frac{GM(4m)}{R^2} = 4F$$

i.e. the force of gravitation becomes 4 times the original value.

68. i. Let the two bodies have masses m_1 and m_2 and they are placed at the same distance R from the centre of the earth. According to the question, if the same force acts on both of them, then

$$F_1 = \frac{GMm_1}{R^2} \dots(i)$$

$$\text{and } F_2 = \frac{GMm_2}{R^2} \dots(ii)$$

$$\text{As, } F_1 = F_2$$

$$\text{Hence, } \frac{GMm_1}{R^2} = \frac{GMm_2}{R^2}$$

So, $m_1 = m_2$, their masses will be the same.

- ii. Mathematically, $g = \frac{GM}{R^2}$

Where, g = acceleration due to gravity

G = universal gravitational constant, M = mass of the earth and R = radius of the earth

- iii. G is known as the universal gravitational constant because its value remains the same all the time everywhere in the universe, applicable to all bodies whether celestial or terrestrial.

69. i. The cube will experience a greater buoyant force in saturated salt solution than in water because density of saturated salt solution is more than the density of water. If each side of the cube is reduced to 4 cm, it will result in reduction in volume of the cube. Hence, the buoyant force experienced by it will reduce in water.

- ii. Buoyant force = weight of displaced water

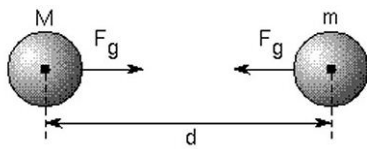
$$= \text{density of water} \times \text{volume of displaced water} \times g$$

$$= 1000 \times \frac{4}{4000} \times 10 \left[\because \text{volume} = \frac{\text{weight}}{\text{density}} \right]$$

$$= 10 \text{ N}$$

70. **Newton's Law of universal gravitation:** Everybody in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

Let us consider two bodies A and B of masses m_1 and m_2 which are separated by a distance d.



Then the force of gravitation (F) acting on the two bodies is given by

$$F \propto m_1 \times m_2 \dots\dots\dots(1)$$

$$\text{and } F \propto \frac{1}{d^2} \dots\dots\dots(2)$$

Combining equations (1) and (2), we get

$$F \propto \frac{m_1 \times m_2}{d^2}$$

$$F = k \frac{m_1 \times m_2}{d^2}$$

Where, k= proportionality constant, known as universal gravitational constant, G having value = $6.67 \times 10^{-11} \text{ Nm}^2 \text{ kg}^{-2}$.

Therefore, $F = G \times \frac{m_1 m_2}{d^2}$, Which is required expression for force of attraction between two bodies.

Here, if the masses m_1 and m_2 of the two bodies are of 1 kg and the distance (d) between them is 1 m, then putting $m_1 = 1 \text{ kg}$, m_2

= 1 kg and $d = 1 \text{ m}$ in the above formula, we get

$$F = G \times \frac{1}{1^2},$$

$$G = F$$

Definition of the gravitational constant G: Gravitational constant, G is numerically equal to the force of gravitation which exists between two bodies of unit masses kept at a unit distance from each other.

71. A stone is dropped from the edge of a roof.

Given, initial velocity $u = 0$

Acceleration $g = 9.8 \text{ m/s}^2$

i. Displacement $= s = 4.9 \text{ m}$

$$\text{We have, } s = ut + \frac{1}{2}gt^2$$

$$4.9 = 0 \times t + \frac{1}{2} \times 9.8 \times t^2$$

$$t^2 = \frac{9.8}{9.8} = 1$$

$$\Rightarrow t = 1 \text{ s}$$

The stone takes 1 s to fall 4.9 m

ii. We have, $v^2 - u^2 = 2as$

$$v^2 - 0^2 = 2 \times 9.8 \times 4.9$$

$$v^2 = 96.04$$

$$\Rightarrow v = \sqrt{96.04} = 9.8 \text{ m/s}$$

At the end of 4.9 m, stone will be moving at a speed of 9.8 m/s

iii. We have, $v^2 - u^2 = 2as$

$$v^2 - 0^2 = 2 \times 9.8 \times 7.9$$

$$v^2 = 154.84$$

$$\Rightarrow v = 12.44 \text{ m/s}$$

The stone will be moving with a speed of 12.44 m/s at the end of 7.9 m.

iv. During the free fall the acceleration produced in a body remains constant.

So, acceleration after 1 s = 9.8 m/s^2

Acceleration after 2 s = 9.8 m/s^2

72. Suppose the mass of the moon is M_m and its radius is R_m . Let a body of mass m is placed on the surface of the moon.

Therefore, the weight of the body on the moon, $W_m = mg' = \frac{GM_m m}{R_m^2} \dots\dots\dots(1)$

Mass of body, m will remain same on earth and moon.

Here, The weight of the same body on the earth's surface, $W_e = mg = \frac{GM_e m}{R_e^2} \dots\dots\dots(2)$

Where M_e = Mass of earth and R_e = Radius of the earth.

Dividing equation (1) by (2), we get

$$\frac{W_m}{W_e} = \frac{M_m}{M_e} \times \frac{R_e^2}{R_m^2} \dots\dots\dots(3)$$

Now, the mass of the earth, $M_e = 6 \times 10^{24}$ kg (Given)

mass of the moon, $M_m = 7.4 \times 10^{22}$ kg (Given)

The radius of the earth, $R_e = 6400$ km and radius of the moon, $R_m = 1740$ km

Now, Putting these values in equation (3), we get

$$\frac{W_m}{W_e} = \frac{7.4 \times 10^{22} \text{ kg}}{6 \times 10^{24} \text{ kg}} \times \left(\frac{6400 \text{ km}}{1740 \text{ km}} \right)^2$$

$$\text{or } \frac{W_m}{W_e} \approx \frac{1}{6} \text{ or } W_m = \frac{W_e}{6} \text{ Hence proved.}$$

Therefore, The weight of the body on the moon is about one-sixth of its weight on the earth.

$$73. F_{\text{gravitation}} = \frac{G \times M_e \times m_o}{r^2}$$

$$= \frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 1}{(6.4 \times 10^6)^2}$$

$$= \frac{6.67 \times 6 \times 10^{-11+24}}{6.4 \times 6.4 \times 10^{12}}$$

$$= \frac{6.67 \times 6}{6.4 \times 6.4} \times 10^{-11+24-12}$$

$$= 0.9770 \times 10 \text{ N} = 9.770 \text{ N}$$

74. i. a. Gravitational force acting on the 50 kg,

$$F = mg = 50 \times 9.8 = 490 \text{ N}$$

b. Gravitational force acting on the 50 kg mass due to jupiter,

$$F_{\text{Jupiter}} = \frac{G \times M_{\text{jupiter}} \times M_{\text{person}}}{(\text{distance of jupiter from the earth})^2}$$

$$F_{\text{Jupiter}} = \frac{6.67 \times 10^{-11} \times 2 \times 10^{27} \times 50}{6.3 \times 10^{11} \times 6.3 \times 10^{11}}$$

$$F_{\text{Jupiter}} = 1.68 \times 10^{-5} \text{ N}$$

c. Gravitational force acting on the 50 kg mass due to saturn

$$F_{\text{saturn}} = \frac{G \times M_{\text{saturn}} \times M_{\text{person}}}{(\text{distance of saturn from the earth})^2}$$

$$F_{\text{saturn}} = \frac{6.67 \times 10^{-11} \times 6 \times 10^{26} \times 50}{1.28 \times 10^{12} \times 1.28 \times 10^{12}}$$

$$F_{\text{saturn}} = 1.12 \times 10^{-6} \text{ N}$$

$$\therefore \text{Total gravitational force due to the Jupiter and the Saturn} = (1.68 \times 10^{-5} + 1.12 \times 10^{-6}) = 1.8 \times 10^{-5} \text{ N}$$

Thus, the combined force due to the planets Jupiter and Saturn (1.8×10^{-5}) N is negligible as compared to the gravitational force i.e. 490 N due to the earth.

ii. We know that g at the equator is less than g at poles (Antarctica). Thus, weight at the equator is less than weight at the pole (Antarctica). A bag of sugar weighs 'w' at a certain place on the equator. If this bag is taken to Antarctica, then it will weigh more due to the greater value of g .

75. Since the weight of any person on the moon is about 1/6 times that on the earth, hence acceleration due to gravity at the moon is 1/6 of that on earth. This means that by applying the same force a person can lift six times heavier objects on the moon than what he could lift on the earth. So, the maximum mass which can be lifted by the same force applied by the person on the moon is $6 \times 15 \text{ kg} = 90 \text{ kg}$.